



Nature-inspired silicone-polythiourethane coatings based on capsaicin analogue for marine biofouling control

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ABSTRACT

Capsaicin analogue, as natural product analogues, have demonstrated significant anti-fouling effects. However, few studies have explored the fluorescence properties of capsaicin in marine biofouling control. A novel capsaicin analogue, *N*-(2-ethoxynaphthalene)-acrylamide (NEA), was successfully synthesized through a Friedel-Crafts alkylation reaction. A series of nature-inspired silicone-polythiourethane coatings based on capsaicin analogue, denoted as PP-NEA-x, were prepared via a thiol-ene click reaction. Due to the polythiourethane matrix, these coatings exhibited excellent mechanical properties. In addition to the fluorescent response of NEA to the UV light in seawater, the typical capsaicin backbone imparts antifouling properties. Owing to the synergistic effects of approach-avoidance mechanism and fluorescence characteristic of NEA, silicone-polythiourethane coatings with fouling-release properties achieved an anti-attachment rate of above 90 % against *Chlorella* and mussel byssus proteins. Furthermore, the coatings demonstrated outstanding environmental compatibility, as confirmed by *Artemia salina* biotoxicity tests. These results suggest that the PP-NEA-x coatings represent a promising alternative to conventional antifouling materials, offering a combination of environmental sustainability, enhanced mechanical performance, and antifouling efficacy.

1. Introduction

The ocean is a highly complex environment characterized by varying water temperature, flow rate, salinity, pH levels, and nutrient concentrations across different regions. This environmental diversity also influences marine organisms, with various fouling species exhibiting preferences for specific surface characteristics of coatings. The attachment of fouling organisms to ships and offshore structures can result in a range of environmental and economic hazards [1], including accelerated metal corrosion, increased hull weight, higher fuel consumption, and elevated greenhouse gas emissions [2,3]. Additionally, biofouling

can facilitate the trans-regional migration of exotic species [4].

Current research in marine antifouling focuses on several key areas: bioactive antifouling agents, self-polishing effects, surface wettability characteristics, photocatalysis, and biomimetic effects [5–11]. In particular, low surface energy (SE) coatings, which utilize surface wettability properties along with modulus of elasticity and surface free energy, reduce biofouling adhesion by leveraging high-speed shear forces generated during ship movement. This approach minimizes the reliance on heavy metal biocides, thereby reducing environmental impact [12–14]. However, these coatings often suffer from poor mechanical properties and weak bonding strength with substrates, which

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can shorten their lifespan. To address this issue, using a polymer with strong adhesion and mechanical properties, such as polythiourethane (PTU), as a matrix and modifying it for low SE is a promising strategy [15,16]. Additionally, their effectiveness may decrease under static conditions due to limited shear forces from low-speed water flow [17]. Therefore, introducing functional groups containing antifoulants or zwitterions can enhance the antifouling properties of these coatings [18–22].

In nature, different corals employ various strategies to achieve antifouling strategies, including (1) the secretion of metabolites with antifouling properties [23]; (2) the evolution of biological surfaces characterized by low SE [24]; (3) the shedding of surface cells, either individually or in patches, to facilitate self-cleaning [25]; (4) the deformation of coral tentacles in response to fluid flow, which generates harmonic motion to enhance antifouling [26]; and (5) the production of fluorescence to inhibit the attachment of fouling organisms [27,28]. Consequently, extensive research has focused on natural or nature-inspired bioactive compounds [29]. Among these, capsaicin, a stimulating vanillamide alkaloid extracted from chili peppers, has been recognized as a potential antifouling agent due to its effectiveness, natural origin, non-toxicity, and ease of environmental absorption [30]. Consequently, capsaicin is recognized as an environmentally friendly antifoulant with significant potential for practical application [31–34]. However, the extraction of capsaicin is limited by factors such as

biomass availability, the concentration of bioactive components, and the technologies used for separation and extraction, which do not sufficiently satisfy the demands of large-scale applications. To overcome these challenges, various capsaicin analogue with functional structures similar to those of natural capsaicin have been synthesized, demonstrating comparable bioactivity and favorable environmental properties [35].

Capsaicin and its analogue possess not only antifouling activity but also ultraviolet fluorescence response. The incorporation of fluorescent-responsive materials into coatings can effectively reduce biofouling deposition on surfaces, providing early-stage prevention of marine biofouling. The antifouling mechanism of fluorescence is based on the fact that low light levels can disrupt the growth and metabolism of marine organisms. Specifically, light affects the physiological activity of cells and interferes with the spatial relationship between chloroplasts and fixed structures within the follicular membrane. This additional illumination prompts fouling organisms to prioritize photosynthesis over attachment [36–40]. Furthermore, fouling organisms adapted to dark environments are negatively affected by fluorescence, impairing their normal physiological activities, including growth, reproduction, gonad maturation, and spawning. Therefore, light may offer a potential strategy for inhibiting fouling by primary organisms [41]. Additionally, UV light, which serves as the excitation wavelength for capsaicin, has a high penetration capacity in seawater. Marine fouling organisms,

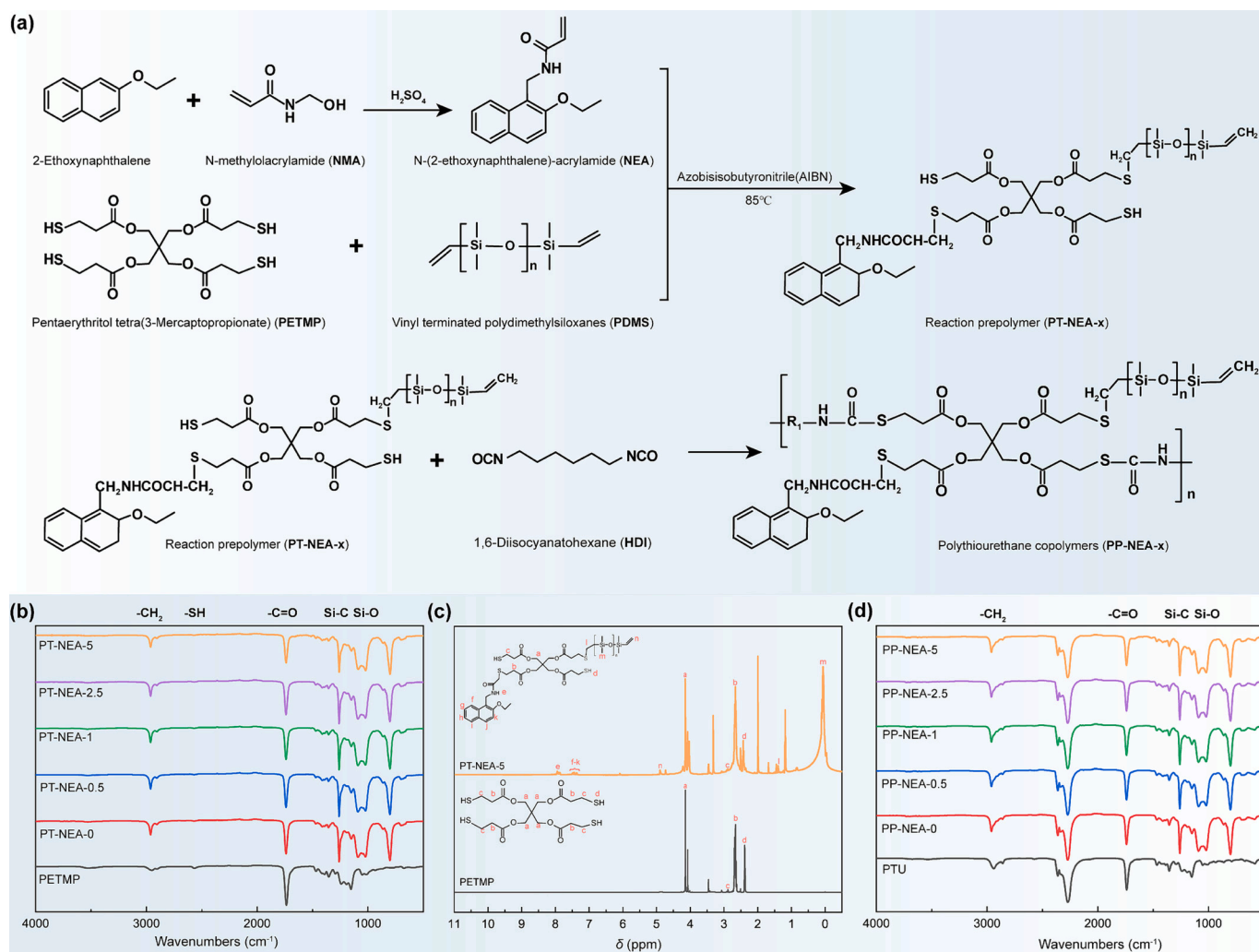


Fig. 1. (a) Synthesis process of nature-inspired silicone-polythiourethane coatings based on capsaicin analogue; (b) FT-IR spectra of PETMP and PT-NEA-x; (c) 1H NMR spectra of PETMP and PT-NEA-5; (d) FT-IR spectra of PTU and PP-NEA-x.

typically active within a few meters of the water's surface, can still be affected by UV-induced fluorescence of capsaicin and its analogue [42,43].

In this study, a nature-inspired silicone-polythiourethane coating based on capsaicin analogue (PP-NEA-*x*) was developed to combine the excellent mechanical durability and adhesive strength of PTU, the fouling release property of PDMS, and the biological approach-avoidance and fluorescence-based antifouling properties of the capsaicin and its analogue structure. First, the novel capsaicin analogue, *N*-(2-ethoxynaphthalene)-acrylamide (NEA) was synthesized via Friedel-Crafts alkylation reaction using an acid catalyst. The NEA synthesis process is shown in Fig. 1a. The NEA structure and vinyl-terminated polydimethylsiloxanes (PDMS) component were then incorporated into a PTU matrix through a thiol-ene click reaction. The fluorescence properties of NEA and PP-NEA-*x* were investigated to assess their impact on the attachment of fouling organisms. The surface properties of the PP-NEA-*x*, including microtopography, roughness, wettability, and adhesion strength to the substrate, were evaluated to ensure their suitability for practical applications in marine environments. PP-NEA-*x* coatings demonstrated excellent performance in inhibiting the attachment of microalgae and mussels and were confirmed to be environmentally friendly through *Artemia salina* biotoxicity test. This innovative antifouling mechanism offers new possibilities for the development and design of environmentally sustainable marine antifouling materials.

2. Materials and methods

2.1. Materials and chemicals

N-methylolacrylamide, petroleum ether, 2-ethoxynaphthalene, PDMS (Mw = 2000) and azobisisobutyronitrile were purchased from Aladdin Bio-Chem Technology Co., Ltd. (Shanghai, China). Ethyl acetate (EA), pentaerythritol tetra(3-mercaptopropionate) (PETMP) and 1,6-diisocyanatohexane (HDI) were obtained from Macklin Biochemical Co., Ltd. (Shanghai, China). Sulfuric acid (H₂SO₄, GR) and ethanol absolute (AR, ≥99.7 %) were purchased from Shanghai Sinopharm Group Chemical Reagent Co., Ltd. (Shanghai, China).

2.2. Sample preparation

2.2.1. Synthesis of NEA

NEA was synthesized via Friedel-Crafts alkylation reaction. To a stirred solution of 2-ethoxynaphthalene (17.2 g, 0.1 mol) and *N*-methylolacrylamide (12.1 g, 0.12 mol) in ethanol (100 mL) was added dropwise a suspension of concentrated sulfuric acid (10 mL) at room temperature. After another 96 h of stirring, the mixture was filtrated and washed with deionized water to get a white precipitate. The crude product was further recrystallized from ethanol to get pure powder.

A Fourier transform infrared spectrometer (FT-IR, Bruker, EQUINOX 55) was used with a spectral scanning range of 400–4000 cm⁻¹ and a spectral resolution of 4.5 cm⁻¹. ¹H NMR spectra were recorded using an AVANCE (600 MHz) instrument (Bruker, Switzerland) in DMSO-*d*₆ as the solvent, with TMS as an internal standard.

IR (KBr) ν : 1242.6, 1463.6, 1545.2, 1617.4, 1654.2, 3062.9, 3284.9 cm⁻¹; ¹H NMR (DMSO-*d*₆, 600 MHz) δ : 1.39 (t, 3H, CH₃), 4.23 (m, 2H, C-CH₂-O), 4.81 (d, 2H, CH₂), 5.56 (m, 1H, CH=C), 6.16 (m, 1H, CH=C), 6.27 (m, 1H, CH=C), 7.38 (m, 1H, Naphth), 7.46 (d, 1H, Naphth), 7.51 (m, 1H, Naphth), 7.88 (d, 1H, Naphth), 7.94 (m, 2H, Naphth), 8.16 (t, 1H, NH).

The structure of NEA is shown in the Supporting Information (Fig. S1). The corresponding IR and ¹H NMR spectra are shown in Figs. S2-S3.

2.2.2. Synthesis of PP-NEA-*x*

Pure polythiourethane can be synthesized through the polycondensation reaction of polyfunctional mercaptoallyl ester and

hexamethylene diisocyanate. For convenience, the synthesized polythiourethane copolymers are designated as PP-NEA-*x* (*x* = 0, 0.5, 1, 2.5, 5), where *x* represents the molar percentage of NEA in the copolymer. The compositions of these copolymers are provided in Table S1. As an example, the preparation of PP-NEA-5 follows these steps: *N*-methylolacrylamide (25 mmol), PDMS (7.8 mmol), and NEA (0.4 mmol) were dissolved in 180 mmol of ethyl acetate, which was then heated to 85 °C. A solution containing 1.77 mmol of initiator azobisisobutyronitrile in ethyl acetate was slowly added to the flask. The reaction was conducted with continuous stirring at 500 rpm using a magnetic stirrer. The heating and stirring process lasted for 3 h, resulting in the formation of the reaction prepolymer (PT-NEA-5). The prepolymer was cooled to room temperature and the ethyl acetate solvent was removed using a rotary evaporator. Subsequently, 41.1 mmol of HDI was added and stirred for 2 h at room temperature, maintaining a stirring speed of 500 rpm to produce the PP-NEA-5 polymer solution. This solution was cast onto glass sheets or Q235 steel plates and cured at room temperature for 48 h, resulting in a nature-inspired silicone-polythiourethane coating based on capsaicin analogue. A schematic diagram of the PP-NEA-*x* synthesis process is shown in Fig. 1a.

2.3. Fluorescence performance assay

2.3.1. UV absorption spectrum

The UV absorption spectrum of NEA was measured by UV-visible spectrophotometer (Persee TU-1900) with chloroform as the background solvent. The excitation wavelength for fluorescence test was determined based on the maximum absorption wavelength observed in the UV spectrum.

2.3.2. Fluorescence spectrum

The powdered sample was utilized as the test substance, and the fluorescence emission spectrum was analyzed using a fluorescence spectrometer (Horiba FluoroMax 4). The excitation wavelength was set to the maximum absorption wavelength previously measured. The spectrum was scanned with a slit width of 5.0 nm, a scanning speed of 200 nm/min, and a scanning range of 200–500 nm.

2.4. Scanning electron microscopy characterization

Surface morphology and elemental distribution of PTU and PP-NEA-*x* coated surfaces were characterized by scanning electron microscopy (SEM) equipped with an energy dispersive spectroscopy (EDS) system (Zeiss Sigma 500). The working distance and accelerating voltage for EDS mapping measurements were 5.9 mm and 15 kV, respectively.

2.5. Surface roughness characterization

The surface morphology and roughness of the coating were characterized using confocal laser scanning microscopy (CLSM, Olympus OLS4000). An area scanning technique was employed to examine the surface morphology. The surface roughness was measured, and the average value of the roughness was calculated to be the final test result.

2.6. Surface wettability characterization

The contact angle of the liquid on the coating surface was measured at room temperature using a drop shape analyzer (KRÜSS DSA100) with a droplet volume of 2 μ L. Deionized water and diiodomethane were used as test liquids [44]. The measurements were performed on seven different areas of each coating under identical conditions, and the average contact angle value was calculated. The surface contact angle (WCA) of the materials was measured before and after two weeks of seawater immersion. The SE of the coatings, including both polar and dispersive components, was determined using the Owens-Wendt-Rabel-Kaelble (OWRK) method. The formula is as follows:

$$\begin{aligned}\gamma_s &= \gamma_s^d + \gamma_s^p \\ \gamma_i(1 + \cos\theta) &= 2\sqrt{\gamma_s^d \gamma_i^d} + 2\sqrt{\gamma_s^p \gamma_i^p}\end{aligned}\quad (1)$$

2.7. Adhesion to substrate test

Adhesion testing was conducted in accordance with ASTM D4514 using a pull-off adhesion tester (PosiTest AT-A Automatic) [45]. The objective was to measure the adhesion strength of PP-NEA-x (with a dry thickness of approximately 300 μm) to a Q235 steel plate. Aluminum spindles, 20 mm in diameter, were affixed to the material's surface using an epoxy resin-based adhesive. The force required to separate the coating from the substrate was measured after the adhesive had dried. The adhesion tester was set to run at a tensile rate of 5 mm/min. The adhesion force recorded during the test reflects the perpendicular force applied to the material surface at a specific rate. Five different points on each coating sample were tested to calculate an average adhesion strength.

2.8. Anti-algae attachment assay

To eliminate contamination, all experimental containers were autoclaved prior to use. *Chlorella* was obtained from the Algae Laboratory at the College of Marine Science and Technology, Zhejiang Ocean University. *Chlorella* solutions in the exponential growth phase were diluted to various concentrations, and absorbance values were measured at the maximum absorption wavelength (660 nm). Standard curves were established for the algal solutions at different absorbances and concentrations, as shown in Fig. S4. The initial biomass of the *Chlorella* solution was adjusted to approximately 3.5×10^5 algal cells per milliliter. The prepared samples (coated glass slides measuring 20×20 mm) were immersed in 10 mL of algal solution. These samples were then placed in a light incubator set to a light intensity of 4000 lx, a temperature of 20 ± 2 °C, and a 12:12 h light-dark cycle. The samples were incubated for 3 days under static conditions. After incubation, the samples were removed and rinsed with deionized water to remove any unattached *Chlorella*. Fluorescence microscopy was used to examine the attachment of algae. The number of attached *Chlorella* was quantified by the fluorescent area observed in fluorescence microscope images. The rate of inhibition of algal colonization was calculated using the formula:

$$T = \frac{T_0 - T_1}{T_0} \times 100\% \quad (2)$$

where T represents the inhibition rate of algal colonization, T_0 is the initial area of algal colonization in the control group, and T_1 is the final area of *Chlorella* colonization in the experimental group. This calculation was repeated three times, and the average value was considered the result, with a confidence value of $p < 0.05$.

2.9. Mussel settlement test

Mussels are considered a large biofouling species that can cause harm to marine facility surfaces. They were selected as the model species for this test. The antifouling effectiveness of PTU and PP-NEA-x was assessed using mussels collected from the nearshore area of Gouqi Island, Zhoushan City, Zhejiang Province. These coatings were randomly placed in an incubator, and twelve healthy mussels were selected and positioned on the coating surfaces. During the experiment, the mussels were allowed to move and regroup. After a four-week experiment period, the mussels were removed from the incubator, and the number of mussel byssus attached to the coated surfaces was recorded.

2.10. Biotoxicity test

Artemia salina, an invertebrate microcrustacean commonly found in

various marine environments, was chosen as the model species for the biotoxicity test. The larval stage of *Artemia salina* is transparency, which facilitates the observation of its physiological and behavioral responses, making it particularly suitable for toxicity assessments and quantitative analyses. Additionally, *Artemia salina* is highly sensitive to a range of toxic substances, including heavy metals and chemical pollutants, enabling effective evaluation of the potential environmental risks. In the experiment, *Artemia salina* was incubated in artificial seawater at 25 °C with a light intensity of 3000–4000 lx for 24 h. Stage I nauplii were then transferred to fresh artificial seawater and cultured for an additional 24 h under the same light conditions. Stage II-III nauplii, which are particularly sensitive to toxic substances due to their unsegmented structure, were subsequently collected for toxicity test. The prepared PTU and PP-NEA-x coatings were immersed in 60 mL of seawater, with 30 *Artemia salina* added to each sample. In the blank control group, only 60 mL of seawater and 30 *Artemia salina* were added. The experiment was conducted in triplicate. After 24 h of incubation at 25 °C in the dark, the mortality of *Artemia salina* was observed under a stereomicroscope. The mortality of *Artemia salina* was determined by whether the nauplii sank to the bottom and remained motionless for 10 s. The mortality rate for each experimental group was then calculated [46].

2.11. Statistical analysis

Statistical analyses of the experimental results were conducted using one-way ANOVA (Tukey test). In all analyses, a p -value of <0.05 was considered statistically significant.

3. Results and discussion

3.1. Chemical composition characterizations

A FT-IR (Bruker, EQUINOX 55) was used with a spectral scanning range of 400–4000 cm^{-1} and a spectral resolution of 4.5 cm^{-1} . ^1H NMR spectra were recorded using an AVANCE (600 MHz) instrument (Bruker, Switzerland) in $\text{DMSO}-d_6$ as the solvent, with TMS as an internal standard. The chemical shifts were reported in parts per million (δ). As shown in Fig. 1b, the peaks at 2960 cm^{-1} and 2860 cm^{-1} correspond to the stretching vibrations of saturated hydrocarbon groups (CH_2 , CH_3). The peak at 2565 cm^{-1} represents the typical SH stretching vibration. The absorption peak at 1740 cm^{-1} is attributed to the C=O vibrational absorption in -COO groups. The Si-CH₃ absorption peaks at 1260 cm^{-1} and 800 cm^{-1} , as well as the Si-O-Si vibrational absorption peaks at 1092 cm^{-1} and 1022 cm^{-1} , are present in the spectra of PT-NEA-x ($x = 0.5, 1, 2.5, 5$) but absent in that of PETMP. ^1H NMR spectra of PETMP and PT-NEA-5 are depicted in Fig. 1c. The peak at 2.4 ppm in both PETMP and PT-NEA-5 is attributed to the thiol proton. The NH signal of NEA appears at approximately 8.1 ppm in the PT-NEA-5 spectrum. A similar change is found in the aromatic region (7.3–7.7 ppm), where the naphthalene ring is present in the PT-NEA-5 spectrum. However, the NH and naphthalene proton signals are absent in the PETMP spectrum. The IR and ^1H NMR spectra confirm the copolymerization of NEA and PDMS components with PETMP. Fig. 1d displays the IR spectra of the copolymer PTU and PP-NEA-x. The complete disappearance of the SH peak in the IR spectra indicates that HDI has fully reacted with the prepolymer. Overall, the formation of the desired product was confirmed by the IR and ^1H NMR spectra of both prepolymer (PT-NEA-x) and copolymer (PP-NEA-x).

3.2. Fluorescence performance

The antifouling ability of NEA is influenced not only by the structure of capsaicin but also by its fluorescence properties. The excitation and emission spectra of NEA are depicted in Fig. 2a. The maximum absorption wavelength is 279 nm, while the maximum emission wavelength is 375 nm, resulting in a Stokes shift of 96 nm. Notably, there is

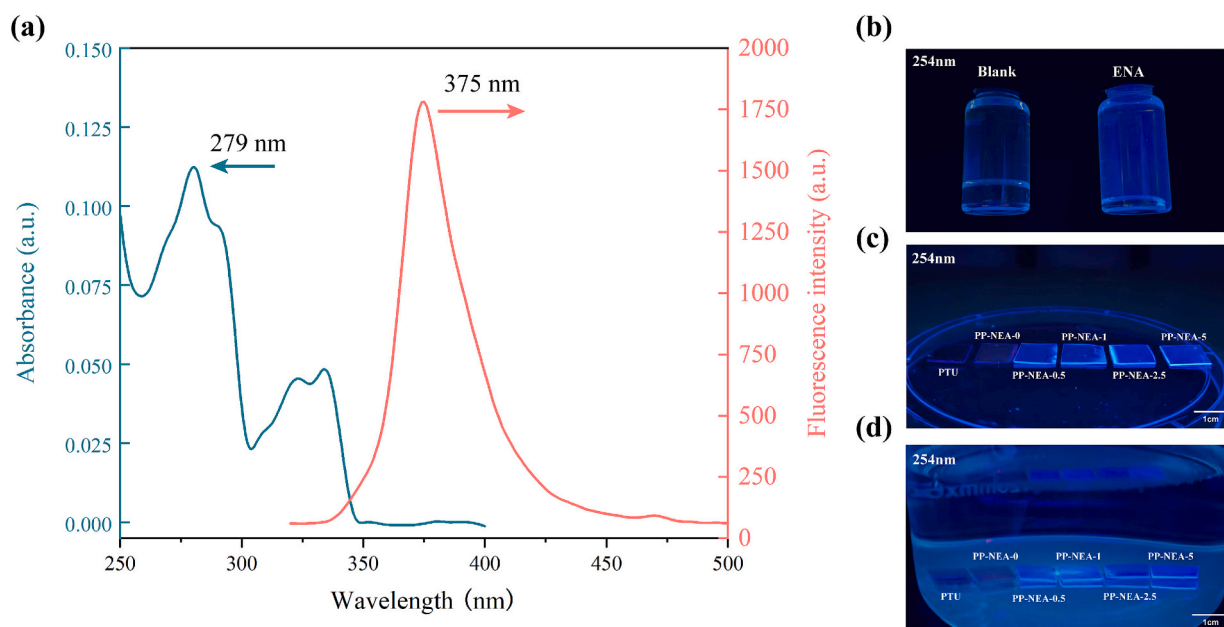


Fig. 2. (a) Excitation and emission spectra of NEA; (b) fluorescence images of ethyl acetate solutions with and without NEA under UV light irradiation; fluorescence images of PTU, PP-NEA-0, and PP-NEA-5 in air (c) and water (d) under UV light irradiation.

minimal overlap between the excitation and emission spectra. This lack of overlap helps mitigate self-absorption, where some of the emitted fluorescence could be reabsorbed during the process. NEA exhibits a large Stokes shift and minimal spectral overlap, reducing interference from autofluorescence and making it less susceptible to fluorescence quenching and self-absorption. The excellent fluorescence property of NEA is closely related to its aromatic structure and substituent group. First, the fluorescence efficiency is directly linked to the extent of conjugation. The polycyclic aromatic hydrocarbon (PAH) in NEA has higher fluorescence efficiency compared to a single benzene ring structure. Second, the introduction of electron-donating groups, such as alkoxy (-OR) and amide (-NHCOR) on the PAH, enhances fluorescence through the formation of a π - π conjugation effect [47]. Furthermore, the position of the substituents is crucial for fluorescence enhancement. Substituents in the neighboring or opposite positions of the aromatic ring enhance fluorescence, while interstitial substitutions tend to inhibit it. Overall, the combination of the greater conjugation effect of PAH, electron-donor groups, and ortho-substitution imparts superior fluorescence properties to NEA.

As shown in Fig. 2b, NEA dissolved in ethyl acetate showed an indigo light under UV irradiation. When exposed to UV light, the PTU and PP-NEA-0 coatings retained their original colors (colorless transparent and milky white, respectively), while the PP-NEA-5 coating emitted an indigo light (Fig. 2c). These results indicate that the coatings introducing the NEA structure maintain the fluorescence properties of NEA. Furthermore, the PP-NEA-x coatings continued to exhibit indigo fluorescence after being immersed in seawater and exposed to UV light (Fig. 2d). This can be attributed to seawater's high transparency to ultraviolet light. Research has demonstrated that in the Western Pacific Ocean, for every meter drop in seawater depth, the intensity of ultraviolet light at 375 nm decreases by only 5 % [43]. The underwater fluorescence image of coatings, along with existing literature, demonstrates that UV attenuation in seawater does not significantly affect the fluorescence performance of coatings in marine antifouling environments.

3.3. Surface topography and roughness performance

The surface morphology and roughness of materials are critical

factors in the attachment of fouling organisms. Rough or uneven surfaces provide more attachment points for biofouling [48–50]. To assess the impact of PDMS and NEA on PP-NEA-x, experimental tests were conducted to analyze the surface morphology and roughness of the coatings. CLSM and SEM images were used to obtain detailed visualizations of the coatings. The SEM images in Fig. 3a illustrate good film-forming properties of PTU and PP-NEA-x. The introduction of PDMS in PP-NEA-0 resulted in some wrinkles on the coating's surface due to the thermodynamic incompatibility between the hard and soft segments in the copolymers. The addition of capsaicin significantly altered the surface morphology of the PP-NEA-x coatings, leading to the formation of a microphase separation structure characterized by circular aggregates in the PP-NEA-0.5, PP-NEA-1, and PP-NEA-2.5. No microphase separation structure was observed in PP-NEA-5. Several factors contribute to this discrepancy. First, π - π interaction between aromatic rings can enhance the attraction between molecules, promoting closer packing of segments containing aromatic groups and facilitating microphase separation. The strong π - π interactions caused by naphthalene group stacking enhanced microphase separation from PP-NEA-0.5 to PP-NEA-1 [51]. Second, microphase separation depends on the flexibility of molecular chains, which allows them to move and spontaneously form different phase domains. However, the introduction of rigid structures can restrict the mobility of polymer segments. When the rigid structure exceeds a certain threshold, the movement of segments is significantly inhibited, preventing the formation of an ordered microphase separation structure [52]. Therefore, while PP-NEA-5 contains the highest NEA content, it shows less pronounced microphase separation. This hypothesis was further validated using EDS. Fig. S5 shows that the S signal from PTU, Si signal from PDMS, and N signal from NEA are dispersed on the surface of the material, satisfying the necessary conditions for the microphase separation structure of circular aggregates formed on the polymer surface.

The surface roughness of the coatings was evaluated using confocal microscopy. The results, shown in Fig. 3b, reveal that the Ra value of the PTU coatings was relatively small, indicating a smooth and flat surface due to the absence of capsaicin. However, the introduction of capsaicin resulted in significant changes in the Ra values of PP-NEA-x, highlighting the notable impact of circular aggregates on the coating roughness. As the degree of phase separation increased, the phase

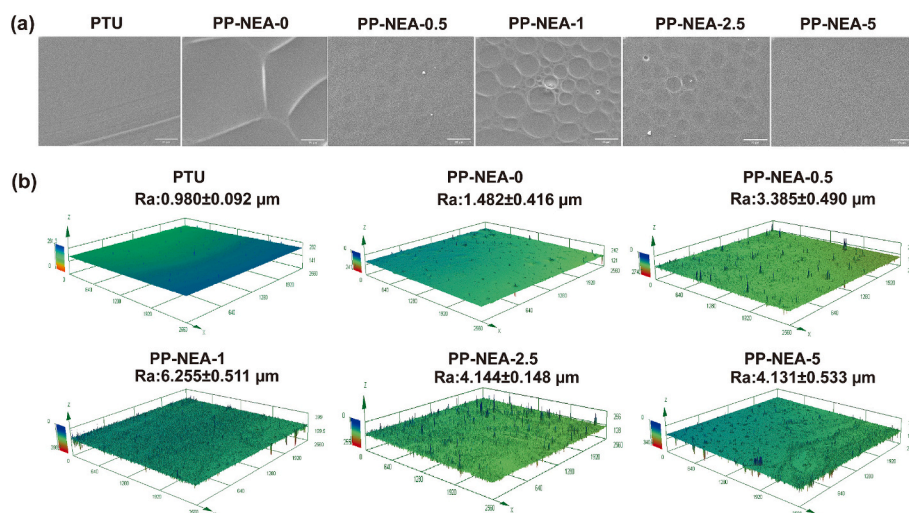


Fig. 3. (a) SEM images of PTU and PP-NEA-x; (b) CLSM images and surface roughness of PTU and PP-NEA-x.

domains formed larger surface undulations, leading to greater surface roughness. Among the coatings, PP-NEA-1 exhibited the highest surface roughness, which can be attributed to the pronounced microphase separation within the copolymer.

3.4. Surface wettability characterization

The surface wettability of PTU and PP-NEA-x coatings was analyzed by investigating WCA and SE before and after 14 days of immersion in artificial seawater (ASW). Raw data are provided in the Supporting Information (Table S2 - S4). As shown in Fig. 4a, the WCA of pure PTU without PDMS is 82.75° . After seawater immersion, the WCA decreases to 59° , indicating increased hydrophilicity. The SE of pure PTU before and after seawater immersion is 35.07 and 44.26 mJ/m^2 (Fig. 4b), respectively. However, these values do not reach the optimal SE range ($20\text{--}25 \text{ mJ/m}^2$) for the best fouling release performance according to the Baier curve, suggesting that pure PTU lacks antifouling properties. The incorporation of PDMS into PP-NEA-0 increases the WCA to 104.10° and decreases the SE to 18.93 mJ/m^2 compared to pure PTU, due to the inherent hydrophobicity of PDMS. Surface roughness is a key factor affecting interfacial wettability, and varying amounts of NEA in PP-NEA-x ($x = 0.5, 1, 2.5, 5$) resulted in different degrees of microphase separation, which in turn influenced the surface roughness. However, in this study, the effect of PDMS on the hydrophobicity of the copolymer far exceeds the effect of surface roughness. As a result, the introduction of the NEA structure did not significantly alter the hydrophobicity of the coatings. SEM and EDS measurements confirmed that the PDMS segments and NEA migrated to the polymer surface, enhancing the water

resistance of the polymer. Consequently, there were no significant changes in WCA and SE after the seawater immersion test. This shows that PP-NEA-x coatings exhibit excellent long-term stability, which is advantageous for their application in real marine environments.

3.5. Adhesion to substrate

In low SE coating systems, fouling organisms are typically removed through shear forces generated by the movement of the ship and gentle mechanical removal devices. Therefore, it is essential to assess the durability of the PP-NEA-x coating by performing adhesion test. The schematic of the adhesion test is shown in Fig. S6. The adhesion strength of coating to the Q235 steel plate substrate is presented in Fig. 4c. The average coating thickness was maintained at approximately $300 \mu\text{m}$. The polar functional groups in the PTU molecular chain, such as the carbamate bond (-NHCOO-), are capable of chemically or physically interacting with the surface polar groups of various substrates (e.g. metal, glass, wood, concrete, etc.). Consequently, the adhesion between PTU and the Q235 steel plate reaches 1.17 MPa , aided by the highly cross-linked structure of the coating.

Hydrogen bonding is a key factor in the adhesion process between the polymer and the substrate. The molecular chains of PTU and PP-NEA-x contain several groups (-COO- and -NHCO-) that facilitate the formation of hydrogen bonds. While the incorporation of NEA reduces the integrity of the cross-linked structure, weakening the polymer's resistance to deformation, the introduction of the amide group in NEA enhances hydrogen bonding between the molecular chains. This results in tighter bonding within the polymer and increased interfacial bonding,

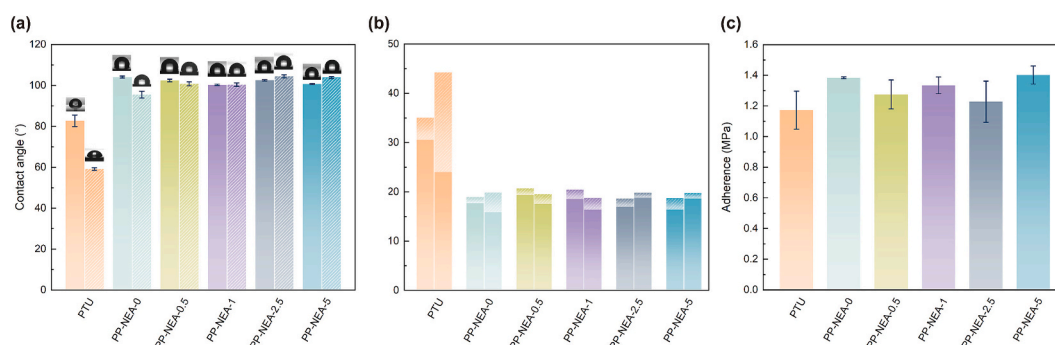


Fig. 4. (a) WCA of PTU and PP-NEA-x before and after 14 days of ASW immersion; (b) SE of PTU and PP-NEA-x before and after 14 days of ASW immersion; (c) adhesion strength of PTU and PP-NEA-x to the substrate Q235 steel plate.

which in turn enhances adhesion. Consequently, the overall adhesion of the polymer remains stable due to the opposing effects of reduced crosslinking and increased hydrogen bonding. As PP-NEA-5 contains the highest amount of NEA structure, it plays a significant role in hydrogen bonding, resulting in the highest adhesion strength. In summary, the adhesion strength of all coatings exceeded 1 MPa, making them suitable for practical marine applications.

3.6. Characterization of antifouling performance

3.6.1. Anti-algae attachment assay

The colonization of marine microalgae is influenced by gravity, van der Waals forces, and electrostatic interactions, which play a critical role in the formation of marine biofouling. To assess the ability to resist algae colonization, an anti-algae attachment experiment was conducted. *Chlorella* colonization on the coated surfaces was observed using fluorescence confocal microscopy, with the results shown in Fig. 5a. As depicted in Fig. 5b, compared to pure PTU, the anti-attachment rates of

Chlorella on PP-NEA-0, PP-NEA-0.5, PP-NEA-1, PP-NEA-2.5, and PP-NEA-5 were 22.6 %, 72.8 %, 85.3 %, 84.3 %, and 93.4 %, respectively. Since the experiment was conducted in a static environment, PP-NEA-0 did not fully exhibit the fouling release properties of low SE coating, resulting in a relatively low anti-attachment rate. However, the introduction of NEA, a capsaicin analogue, significantly improved the algal inhibition performance (from 22.6 % to over 70 %). The higher the NEA content in the coating, the more effective it was in preventing *Chlorella* colonization. This can be attributed to the strong stimulatory effects of capsaicin and its analogue, which interfere with the physiological processes of marine organisms by affecting their nerve receptors, blocking normal signal transmission, and ultimately preventing attachment [53,54]. Additionally, the self-fluorescence properties of PP-NEA-x further contribute to the anti-algal adhesion behavior. In highly transparent seawater, where UV rays can penetrate up to 20 m below sea level and stimulate these compounds to fluoresce. Due to the restricted intramolecular rotation within the solid-state coating, energy is consumed from the radiation channel, resulting in enhanced

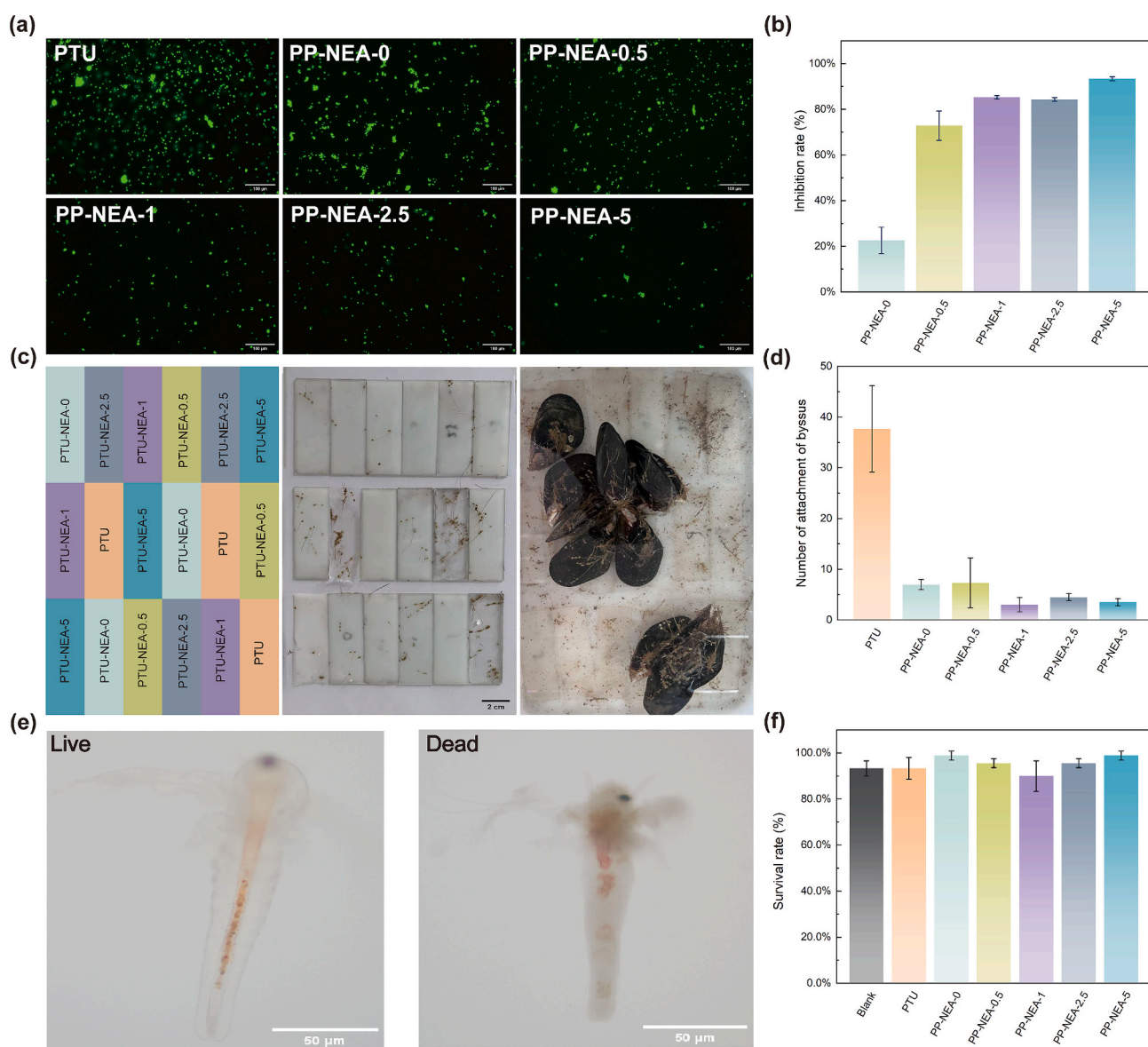


Fig. 5. (a) Fluorescence microscopy images of the PTU and PP-NEA-x surface after 48 h immersion in *Chlorella*; (b) colonization inhibition rate of PTU and PP-NEA-x against *Chlorella*; (c) distribution of randomly dispersed samples, byssus deposition on samples, and images of mussels attached to samples after four weeks; (d) quantification of byssus deposition on PTU and PP-NEA-x surfaces; (e) morphological differences between live and dead *Artemia salina*; (f) survival rate of *Artemia salina* after 24 h of dark incubation in seawater containing PTU and PP-NEA-x.

fluorescence emission, thereby strengthening the fluorescence-based antifouling effect [55,56]. In conclusion, the incorporation of NEA as an antifouling monomer into silicone-polythiourethane significantly reduces the adhesion of fouling organisms to the material surface.

3.6.2. Mussel settlement test

The impact of large fouling organisms, especially hard-shelled species, on marine engineering equipment is substantial. Mussels, which attach firmly to various surfaces by secreting adhesive pedicle filaments, are highly adaptable to complex environments. For this reason, they were selected as the experimental subjects for large-scale fouling bioassay [57,58]. At the onset of the experiment, mussels were evenly distributed on the surface of randomly arranged samples. Using their feet, the mussels assessed the surface properties (e.g., roughness, wettability, chemical composition) and chose suitable attachment sites. As shown in Fig. 5c, mussels deposited byssus proteins on nearly all surfaces, with the highest concentration on the PTU surface and fewer proteins on the PP-NEA-x surfaces. Specifically, the byssus proteins on the PP-NEA-5 surface were only 9 % of those found on the PTU surface. This lower concentration suggests that mussels had fewer attachment points, making them more susceptible to detachment under external forces, such as water flow or mechanical disturbances. These results suggest that highly hydrophobic surfaces, which resist wetting, hinder mussel attachment. Additionally, low SE coatings impede the formation of stable physical or chemical bonds between the mussels and the surface. After 4 weeks of attempting to colonize, mussels preferentially attached to the PTU surfaces, container walls, and PP-NEA-0, where they exhibited clustered growth. Besides the hydrophobic and SE characteristics of the coating, the NEA structure within the molecular chain utilized biological approach-avoidance properties to keep the mussels away from its surface. In summary, PP-NEA-x coating effectively reduces mussel attachment and aggregation through its hydrophobicity, low SE and active NEA structure.

3.6.3. Biotoxicity test

Artemia salina is frequently used as an indicator organism in biotoxicity tests to evaluate the safety and ecological impact of antifouling coatings in aquatic environments. Assessing the toxicity of coatings to *Artemia salina* provides insights into the potential hazards these

materials may pose to aquatic organisms [59,60]. The morphology of *Artemia salina* under different conditions following exposure to seawater is shown in Fig. 5e. Healthy individuals exhibit a transparent body with visible internal organs and tissue structures, a straight carapace, and normal limb movement. In contrast, dead *Artemia salina* loses transparency, appears grayish-white, exhibits stiff bodies, and no longer swims. The results of the toxicity assay for *Artemia salina* exposed to seawater containing antifouling coatings with varying NEA concentrations are depicted in Fig. 5f. The survival rate of *Artemia salina* in the blank group, simulating a natural marine environment, was 93.33 %. In seawater containing PP-NEA-x ($x = 0.5, 1, 2.5, 5$) coatings, survival rates were approximately 90 %. Considering possible physical harm during the experiment, these results suggest that NEA-containing antifouling coatings do not exhibit significant toxicity to non-target aquatic organisms like *Artemia salina*. Furthermore, this indirectly confirms that NEA inhibits organism attachment through physical avoidance rather than lethal effects. These findings provide theoretical support for the environmental friendliness of PP-NEA-x coatings.

3.7. Antifouling mechanism

The antifouling mechanism of nature-inspired silicone-polythiourethane coatings based on capsaicin analogue is illustrated in Fig. 6. When PDMS is incorporated into the polythiourethane matrix, the siloxane chain segments migrate to the coating surface, creating a microphase-separated microstructure. This surface morphology endows the material with significant hydrophobicity and low SE, effectively reducing microorganism adhesion. Upon with diatoms, bacteria, and other microorganisms, the NEA components in the coating structure function as biomimetic antifouling agents, preventing surface adhesion through biological approach-avoidance properties [38]. Additionally, the aromatic ring structure and amide groups in NEA contribute to the formation of resonance structures, facilitating intramolecular electron transfer. During daylight exposure to UV radiation, the NEA structure absorbs energy and emits fluorescence, redirecting fouling organisms toward photosynthesis rather than attachment, thereby enhancing the coating's antifouling performance [42,61,62]. In summary, the superior antifouling performance is attributed to the synergistic effects of the approach-avoidance and fluorescence-based anti-biofouling properties

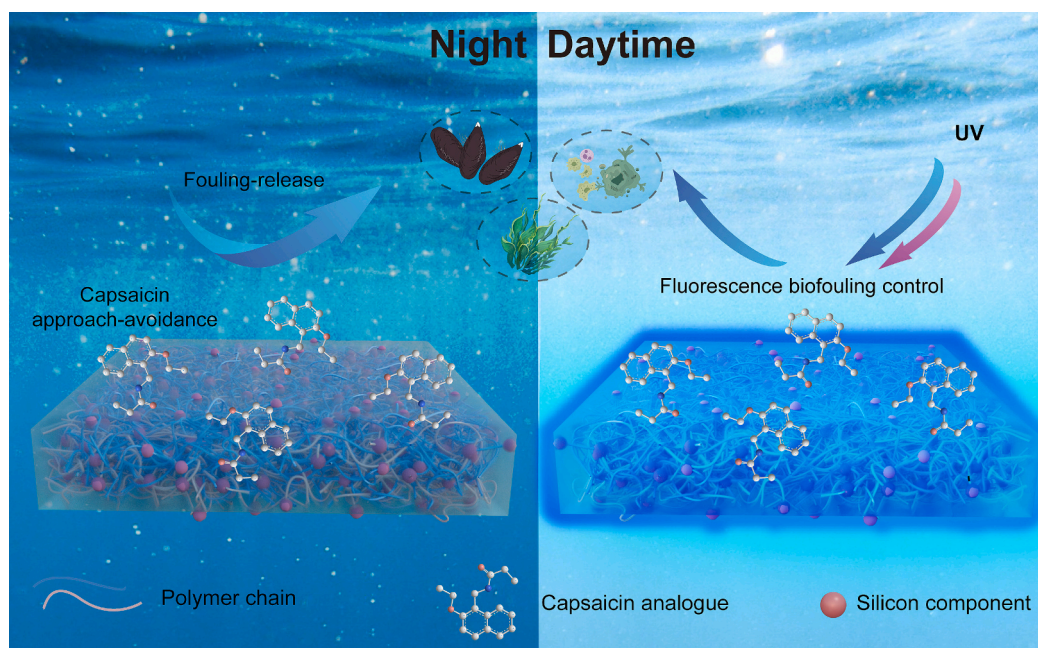


Fig. 6. Antifouling mechanism of nature-inspired silicone-polythiourethane coatings based on capsaicin analogue.

of NEA structure and the fouling-release characteristics imparted by the coating's low SE.

4. Conclusion

In conclusion, nature-inspired silicone-polythiourethane coatings based on capsaicin analogue were successfully prepared using a thiol-ene click reaction. The coatings demonstrated a novel synergistic anti-fouling strategy, combining the approach-avoidance and fluorescence anti-biofouling effects of NEA structure with the fouling-release property of low SE. Under UV light exposure, PP-NEA-x coatings exhibited indigo fluorescence when immersed in seawater. The microphase separation structure of circular aggregates formed on the partial coating surface due to the thermodynamic incompatibility between the hard and soft segments in copolymers and π - π interaction within the NEA structure. PP-NEA-x coatings maintained hydrophobicity and low SE properties after seawater immersion and exhibited excellent adhesion strength, exceeding 1 MPa, making them suitable for marine applications. The coatings also showed significant anti-attachment effects against algae and mussels, with maximum inhibition rates exceeding 90 %, while posing no significant toxicity to non-target aquatic organisms such as *Artemia salina*. The combination of mechanical durability, adhesive strength, environmental friendliness, and antifouling performance highlights the potential of nature-inspired silicone-polythiourethane coatings based on capsaicin analogue for marine applications.

CRedit authorship contribution statement

Kang Feng: Writing – original draft, Data curation, Conceptualization. **Sheng Li:** Writing – original draft, Data curation. **Najla AlMasoud:** Writing – original draft, Data curation. **Jiaying Li:** Writing – original draft, Data curation. **Yue Ma:** Writing – original draft, Data curation. **Congying Yin:** Writing – original draft, Formal analysis. **Zeyu Jie:** Writing – original draft, Formal analysis. **Yiding Wang:** Writing – original draft. **Taghrid S. Alomar:** Writing – review & editing, Formal analysis. **Xia Li:** Writing – review & editing. **Haoyi Qiu:** Writing – review & editing. **Zeinhom M. El-Bahy:** Writing – review & editing, Formal analysis. **Yan Li:** Writing – review & editing, Project administration, Funding acquisition. **Xingtao Xu:** Writing – review & editing, Supervision, Funding acquisition. **Yusuke Yamauchi:** Writing – review & editing, Supervision. **Liangmin Yu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2025.164362>.

Data availability

Data will be made available on request.

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